

SE446: Big Data Engineering

Review Questions — Weeks 07, 08, 09

Apache Spark & Spark MLlib

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Student Version — Questions Only

Instructions:

- This document contains **54 review questions** covering Apache Spark RDDs, DataFrames, Spark SQL, cluster tuning, and Spark MLlib.
- **Section A:** 16 Conceptual Questions — test your understanding of core concepts.
- **Section B:** 10 Critical Thinking Questions — apply reasoning to realistic scenarios.
- **Section C:** 28 Problem-Solving Questions — write code and solve engineering problems.
- Write your answers in the spaces provided below each question.
- Use Python/PySpark syntax for all code questions.

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1 Section A: General Review / Conceptual Questions (16 Questions)

Question A1

What does RDD stand for? List and briefly explain its three key properties.

Space for your answer:

Question A2

Explain the difference between a **transformation** and an **action** in Spark. Give two examples of each.

Space for your answer:

Question A3

Why is Apache Spark significantly faster than MapReduce for iterative algorithms such as machine learning?

Space for your answer:

Question A4

What is the role of each of the following Spark components?

- (a) Driver Program
- (b) Cluster Manager
- (c) Executor
- (d) Task

Space for your answer:

Question A5

What is the difference between `SparkContext` and `SparkSession`? When do you use each?

Space for your answer:

Question A6

List three advantages of DataFrames over RDDs.

Space for your answer:

Question A7

Name and explain three optimizations that the Catalyst optimizer applies to DataFrame queries.

Space for your answer:

Question A8

What is the difference between `createOrReplaceTempView` and `createOrReplaceGlobalTempView`?

Space for your answer:

Question A9

Compare **client** and **cluster** deploy modes in `spark-submit`. When should you use each?

Space for your answer:

Question A10

Explain what happens when you call `df.cache()`. Does caching happen immediately?

Space for your answer:

Question A11

What is the difference between `repartition(n)` and `coalesce(n)`?

Space for your answer:

Question A12

What is a broadcast join? When should you use it?

Space for your answer:

Question A13

In Spark MLlib, what is the difference between a **Transformer** and an **Estimator**?

Space for your answer:

Question A14

Why does Spark MLlib require all features to be in a single vector column? What tool creates this vector?

Space for your answer:

Question A15

When should you use `StringIndexer` alone vs. `StringIndexer + OneHotEncoder`?

Space for your answer:

Question A16

What are the four file formats discussed in the course? Which one is recommended for Spark-to-Spark workflows and why?

Space for your answer:

2 Section B: Critical Thinking Questions (10 Questions)

Question B1

A student writes RDD code and an equivalent DataFrame query. Both produce the same result, but the DataFrame version runs 3x faster. Explain why the Catalyst optimizer cannot help the RDD version.

```
1 # RDD version
2 rdd.filter(lambda x: x != header) \
3     .map(lambda x: x.split(",")) \
4     .filter(lambda f: f[5] == "THEFT") \
5     .map(lambda f: (f[5], 1)) \
6     .reduceByKey(lambda a, b: a + b)
7
8 # DataFrame version
9 df.filter(col("Primary Type") == "THEFT") \
10    .groupBy("Primary Type").count()
```

Space for your answer:

Question B2

A student's Spark job on a 4-core cluster takes 3 minutes. They check the Spark Web UI and see 200 tasks for a `groupBy` stage, with most tasks completing in 1ms. What is the problem, and how do they fix it?

Space for your answer:

Question B3

The Chicago Crimes dataset has approximately 85% of records with `Arrest = False`. A student trains a model that achieves 85% accuracy and concludes the model works well. What is wrong with this reasoning?

Space for your answer:

Question B4

A student uses `inferSchema=True` when loading a large CSV file and notices the job takes twice as long as expected. Explain why and suggest a better approach.

Space for your answer:

Question B5

A student writes the following code. Identify the **data leakage** problem and explain how to fix it using a Pipeline.

```
1 indexer = StringIndexer(inputCol="Primary Type",
2   outputCol="crime_index")
3 indexer_model = indexer.fit(df)           # fit on FULL data
4 df = indexer_model.transform(df)
5 train_df, test_df = df.randomSplit([0.8, 0.2])
6 model = rf.fit(train_df)
7 predictions = model.transform(test_df)
```

Space for your answer:

Question B6

You call `df.cache()` and then immediately run `df.count()`. You notice the first `df.count()` is not faster than without caching. Why? When does the speedup appear?

Space for your answer:

Question B7

Under what circumstances would you choose scikit-learn over Spark MLlib for a machine learning task?

Space for your answer:

Question B8

A student broadcasts a 500 MB table during a join. What will happen, and why?

Space for your answer:

Question B9

A GBT model achieves an AUC of 0.92 on the Chicago crimes dataset, while Logistic Regression achieves only 0.72. What does this large gap tell you about the nature of the data?

Space for your answer:

Question B10

Your Spark job's Stages tab shows one task taking 45 seconds while all other tasks in the same stage complete in 3 seconds. Diagnose the problem and suggest two possible fixes.

Space for your answer:

3 Section C: Problem-Solving / Engineering Questions (28 Questions)

3.1 Part 1 — RDD Programming (C1–C5)

Question C1

Given the Chicago Crimes CSV file on HDFS, write RDD code to:

- (1) Load the file with `sc.textFile()`
- (2) Skip the header line
- (3) Count the total number of crime records (excluding the header)

Space for your answer:

Write your code in the space above.

Question C2

Using the RDD API, write code to find the **top 5 crime types by frequency**. Assume the CSV has columns: ID, Date, Block, Primary Type, Description, ... where Primary Type is at index 5 (0-based).

Space for your answer:

Write your code in the space above.

Question C3

Using the RDD API, write code to compute the **arrest rate** (percentage of crimes that led to arrest). Assume Arrest is at column index 8 and has values "true" or "false".

Space for your answer:

Write your code in the space above.

Question C4

Using the RDD API, write code to count the number of crimes per district, and return the results as a dictionary to the driver.

Space for your answer:

Write your code in the space above.

Question C5

Identify the **transformations** and **actions** in the following RDD code. For each transformation, state whether it is **narrow** (no shuffle) or **wide** (shuffle).

```
1 rdd = sc.textFile("hdfs:///data/crimes.csv")
2 rdd2 = rdd.filter(lambda x: x != header)
3 rdd3 = rdd2.map(lambda x: x.split(","))
4 rdd4 = rdd3.map(lambda f: (f[5], 1))
5 rdd5 = rdd4.reduceByKey(lambda a, b: a + b)
6 rdd6 = rdd5.sortBy(lambda x: x[1], ascending=False)
7 result = rdd6.collect()
```

Space for your answer:

Write your code in the space above.

3.2 Part 2 — DataFrame Operations (C6–C12)

Question C6

Write DataFrame code to:

- (1) Load the CSV with header and schema inference
- (2) Select only columns `Primary Type`, `District`, `Arrest`
- (3) Filter for crimes where `Arrest` is `True`
- (4) Show the first 10 rows

Space for your answer:

Write your code in the space above.

Question C7

Write DataFrame code to add a new column called `"ArrestLabel"` that contains `"Arrested"` when `Arrest` is `True` and `"Not Arrested"` otherwise. Then show 5 rows with columns `Primary Type`, `District`, `ArrestLabel`.

Space for your answer:

Write your code in the space above.

Question C8

Write DataFrame code to compute multiple aggregations at once: for each `District`, compute the total number of crimes and the arrest rate (average of the `Arrest` boolean cast to integer). Sort by district number.

Space for your answer:

Write your code in the space above.

Question C9

Write DataFrame code to perform an **inner join** between the crimes DataFrame and this small lookup DataFrame, then show the result with district names instead of IDs:

```
1 district_df = spark.createDataFrame([
2     (1, "Central"), (2, "Wentworth"),
3     (8, "Chicago Lawn"), (11, "Harrison"), (12, "Near West")
4 ], ["DistrictID", "DistrictName"])
```

Space for your answer:

Write your code in the space above.

Question C10

Write DataFrame code to:

- (1) Rename the column "Primary Type" to "CrimeType"
- (2) Drop the column "Block"
- (3) Print the schema to verify the changes

Space for your answer:

Write your code in the space above.

Question C11

Write DataFrame code to inspect a DataFrame in three different ways:

- (a) Print the schema
- (b) Show descriptive statistics

(c) Check the number of partitions

Space for your answer:

Write your code in the space above.

Question C12

Write DataFrame code to save the result of a `groupBy` aggregation to HDFS in Parquet format, partitioned by the `District` column, using overwrite mode.

Space for your answer:

Write your code in the space above.

3.3 Part 3 — Spark SQL Queries (C13–C17)

Question C13

Register the crimes DataFrame as a temporary view and write a Spark SQL query to find the **top 10 crime types by count**.

Space for your answer:

Write your code in the space above.

Question C14

Write a Spark SQL query to compute the **arrest rate** (as a percentage) for each crime type, showing only types with more than 50 occurrences, sorted by arrest rate descending.

Space for your answer:

Write your code in the space above.

Question C15

Write a Spark SQL query to find the **top 5 districts** with the highest total crime count.

Space for your answer:

Write your code in the space above.

Question C16

Write a Spark SQL query to show the **number of crimes per year**, sorted chronologically.

Space for your answer:

Write your code in the space above.

Question C17

Write a Spark SQL query that uses a **subquery** to find all crime types whose total count is above the average count across all crime types.

Space for your answer:

Write your code in the space above.

3.4 Part 4 — Three-Way Comparison (C18–C19)

Question C18

Write the solution in **all three APIs** (RDD, DataFrame API, and Spark SQL) for:

Task: Find all crimes of type "NARCOTICS" in District 11, and count them.

Space for your answer:

Write your code in the space above.

Question C19

Write the solution in **all three APIs** for:

Task: Count the number of domestic vs non-domestic crimes, and show both counts.

Space for your answer:

Write your code in the space above.

3.5 Part 5 — Cluster Tuning, MLlib & Evaluation (C20–C28)

Question C20

Write PySpark code to:

- (1) Create a SparkSession
- (2) Load `chicago_crimes.csv` from HDFS
- (3) Count crimes per `Primary Type`, sorted descending
- (4) Show top 10
- (5) Write result to HDFS as Parquet

Space for your answer:

Write your code in the space above.

Question C21

Write the query from C20 using Spark SQL. Explain why both versions produce the exact same execution plan.

Space for your answer:

Write your code in the space above.

Question C22

Given this `explain(True)` output, identify (a) which optimization was applied and (b) how many columns are actually read:

```
== Optimized Logical Plan ==
Aggregate [Primary Type], [Primary Type, count(1) AS count]
+- Project [Primary Type]
   +- Relation[ID, Date, Primary Type, District, Arrest, ...] csv

== Physical Plan ==
HashAggregate(keys=[Primary Type], functions=[count(1)])
+- Exchange hashpartitioning(Primary Type, 200)
   +- HashAggregate(keys=[Primary Type],
                     functions=[partial_count(1)])
      +- FileScan csv [Primary Type]
```

Space for your answer:

Question C23

Write the correct `spark-submit` command for:

- A script `crime_analysis.py` that takes 45 minutes
- Your laptop may lose WiFi
- 2-worker YARN cluster, each with 2 cores and 2 GB RAM
- Shuffle partitions set to 8

Space for your answer:

Write your code in the space above.

Question C24

A student runs 5 actions without caching. How many HDFS reads occur? Rewrite the code with caching and state the new count.

```
1 df = spark.read.csv("hdfs:///data/crimes.csv",
2                       header=True, inferSchema=True)
3 print(df.count())
4 df.groupBy("Primary Type").count().show()
5 df.filter(col("Arrest") == True).count()
6 df.groupBy("District").count().show()
7 df.select("Year").distinct().show()
```

Space for your answer:

Write your code in the space above.

Question C25

Given a cluster with 2 workers, each having 2 cores (4 total):

- (a) What is the ideal partition count?
- (b) What should `shuffle.partitions` be set to?
- (c) What is the problem with 1 partition?
- (d) What is the problem with 200 partitions?

Space for your answer:

Question C26

Write a complete PySpark ML pipeline:

- (1) Encode Primary Type with StringIndexer

- (2) Encode `Domestic` with `StringIndexer`
- (3) Assemble features
- (4) Train `RandomForest` with 100 trees, max depth 5
- (5) Evaluate with AUC-ROC and F1

Space for your answer:

Write your code in the space above.

Question C27

Two workers, weights $\mathbf{w} = [0.5, 0.3]$, learning rate $\eta = 0.1$. Local gradients:

- Worker 1: $[-0.045, 0.145]$
 - Worker 2: $[0.170, 0.330]$
- (a) Compute the aggregated gradient
 - (b) Compute the updated weights
 - (c) What travels over the network?

Space for your answer:

Question C28

Given the following confusion matrix:

	Predicted Negative	Predicted Positive
Actual Negative	TN = 6800	FP = 350
Actual Positive	FN = 420	TP = 1430

- (a) Calculate Accuracy, Precision, Recall, and F1 Score
- (b) Is high FP or high FN worse for crime prediction?
- (c) Which metric should you prioritize?

Space for your answer:

— End of Review Questions —